Revenue & Profitability Problem Statement Brief

Year-over-year financial performance review: PayBridge NG

Prepared by	Prepared for	Review period	Status
Data Analytics	**Executive Leadership**	**FY2025 vs FY2024**	**Draft for approval**

Executive summary

PayBridge NG is a Nigerian payments company processing merchant transactions across POS, web checkout, transfers and bill payments. Management reports healthy top-line growth, yet profit is not keeping pace. This brief defines the problem, the decision it supports, the questions to answer, the KPIs to track and the hypotheses to test, so the analysis is structured before any data is examined.

> **Core question:** Where does PayBridge NG's revenue growth stop converting into profit, and what should management change first?

1 Business problem

In the management scenario, revenue grew strongly year over year while net profit grew only marginally. Finance has no consistent view of *why*, because reporting is built around revenue totals and does not separate growth by channel, merchant, region or cost.

- **Growth without margin.** Net margin is compressing: each additional naira of revenue is retaining less profit.
- **Opaque cost-to-serve.** Processing, switching and bank charges are reported in aggregate, so loss-making segments are hidden inside healthy averages.
- **Unknown dependency.** There is no tracked measure of how much revenue rests on a small number of merchants.
- **Silent revenue loss.** Failed transactions can generate cost with no fee, but this loss is not quantified.
- **Slow, manual reporting.** Monthly performance packs are assembled by hand, delaying decisions.

Note: figures in the scenario are illustrative and will be reproduced through a synthetic dataset built to mirror realistic payments economics.

2 Business objective

Management must decide **where to focus the next 12 months of commercial and cost effort** to restore margin without sacrificing growth. The analysis should enable three specific decisions:

1. Which channels, products and merchant segments to grow, reprice or deprioritise.
2. Which cost lines to renegotiate or automate first, and the expected naira impact.
3. Which regions justify further investment and which are sub-scale.

Deliverable: an interactive Revenue & Profitability dashboard with quantified recommendations, supported by a documented data model and metric definitions.

3 Stakeholders

Stakeholder	Interest	What they need from the analysis
Chief Executive Officer	Sustainable, profitable growth	One-page view of growth quality and top priorities
Chief Financial Officer	Margin, cost control, forecast accuracy	Reconciled revenue, cost and profit by segment, with YoY variance
Head of Product	Product-line performance	Contribution and margin by product and channel
Head of Sales / Merchant Success	Merchant growth and retention	Merchant profitability, concentration and pricing exposure
Head of Operations	Reliability and processing cost	Failed-transaction impact and cost per transaction
Regional Managers	Local targets	City and state performance against prior year
Risk & Compliance	Data integrity, concentration risk	Auditable metrics and documented assumptions

4 Analytical questions

Growth quality

- How much of revenue growth came from volume, from price (take rate) and from mix?
- Is profit growing slower than revenue, and in which months did the gap open?

Segment profitability

- Which channels and products earn the highest and lowest net margin per transaction?
- Which merchants generate high volume but thin or negative margin?

Cost and efficiency

- Is cost per transaction rising faster than fee revenue per transaction?
- Which cost lines (processing, switching, bank charges) drive the margin change?

Risk and dependency

- What share of net revenue comes from the top 10 merchants, and how has it changed?
- How much revenue is lost to failed transactions, and where do failures cluster?

Geography and timing

- Which states and cities are growing, flat or declining, and are they profitable?
- How do seasonality and peak periods affect volume, cost and margin?

5 Proposed KPIs

KPI	Definition	Why it matters
Fee revenue	Total fees charged to merchants	Headline growth measure
Net revenue (profit)	Fee revenue minus processing, switching and bank costs	True earnings from transactions
Net margin %	Net revenue ÷ fee revenue	Quality of growth
Take rate	Fee revenue ÷ transaction value	Pricing power and discount leakage
Cost per transaction	Total cost ÷ transaction count	Cost-to-serve efficiency

KPI	Definition	Why it matters
Transaction count and value	Number and naira value of successful transactions	Volume driver
Revenue concentration	Top 10 merchants' share of net revenue	Dependency risk
Failed-transaction loss	Estimated fees forgone on failed transactions	Recoverable revenue
Active merchants and revenue per merchant	Merchants with at least one transaction in period; fee revenue ÷ active merchants	Breadth versus depth of growth
YoY and YTD variance	Current period vs same period prior year, in naira and %	Core comparison across all KPIs

6 Initial hypotheses

These are informed guesses made before seeing the data. Each is written to be disproved.

Hypothesis	Reasoning	How we test it
H1: Mix shift	Growth is concentrated in a lower-margin channel, diluting overall margin.	Margin and revenue share by channel, FY2024 vs FY2025; mix-effect decomposition
H2: Cost inflation	Per-transaction processing and switching costs rose faster than fees charged.	Cost per transaction vs take rate, monthly trend
H3: Discount leakage	Large merchants negotiated lower rates, so volume grew but take rate fell.	Take rate by merchant tier; margin of top 10 vs the rest
H4: Failed-transaction drag	Failures incur cost without fees, concentrated in specific channels or peak hours.	Success rate by channel and hour; fees forgone
H5: Sub-scale expansion	Newer cities add revenue but sit below break-even margin.	Net margin and cost per transaction by city
H6: Peak-season squeeze	Q4 volume spikes lift cost faster than revenue.	Monthly margin against transaction volume

Working view: H1 and H2 are the most probable primary causes, with H3 amplifying them. If the data supports this, the first recommendation will concern channel pricing and cost renegotiation.

7 Scope, assumptions and next steps

- **Data:** a synthetic dataset of about 24 months of transactions with merchant, product, channel, location and cost-rate tables, built to realistic Nigerian payments patterns.
- **Currency:** Nigerian naira (NGN). Taxes, FX effects and capital costs are out of scope.
- **Metric definitions** in Section 5 are the single source of truth for SQL and DAX.
- **Next steps:** approve this brief, design the data model, generate the dataset, build the SQL layer, then the Power BI report and recommendations.

PayBridge NG is a fictional company created for a portfolio analytics project. All scenario figures are illustrative.